

Data Preparation

Unlabelled WLT-UP Dataset

CLAHE

Location based Splitting

Test

Train

Validation

Test set

Model Evaluation

Finetuned Model

Detection Level
Evaluation

Bounding Box
Evaluation

Transfer Learning

Supplementary Dataset

Domain Adaptation
and Pretraining

Foundational Model
YOLO, Faster R-CNN, and
RT-DETR

Finetuning
on WLT-UP
dataset

Phase 1: Freeze
Backbone

Phase 2: Global
Finetuning

Finetuned
models

Initial Sampling and Annotation

Validation
set

Train set

Active Learning Loop

Inference on
Unlabeled Train set

DCUS

CCMS and k-means++
Clustering

Human Annotation

